



Original Research

Identifying EEG biomarkers of fatigue and brain fog in long COVID: insights from rest and movement-related brain activity

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ABSTRACT

Background: Fatigue and brain fog are among the most reported symptoms in Long COVID (LC), yet their underlying neural mechanisms remain unclear. In this context, we aimed to identify neural correlates associated with LC symptoms.

Methods: Electroencephalography (EEG) was used to quantify brain oscillatory activity during sustained handgrips and rest in the left central region overlying the primary hand motor (C3) and somatosensory areas (C5). A sample of twenty LC participants reporting fatigue, brain fog, and post-exertional malaise (PEM) was analyzed, along with 20 age- and sex-matched controls. All participants were right-handed, and EEG activity was recorded on the dominant left hemisphere over motor and somatosensory areas. The main LC symptoms were assessed using the Fatigue Severity Scale (FSS), the Cognitive Failures Questionnaire (CFQ), and the DePaul Symptom Questionnaire for PEM (DSQ-PEM).

Results: LC participants showed a smaller post-movement beta rebound (PMBR) during handgrips in the left central region (electrode C3, $p = 0.02$) compared to controls, which negatively correlated with cognitive symptoms ($r = -0.35$, $p = 0.02$). They also showed a slower alpha central frequency (CF) at rest (electrode C5, $p = 0.003$), which correlated with fatigue ($r = -0.264$, $p = 0.019$).

Conclusions: Both the lower PMBR and slower alpha CF suggest impaired inhibitory control and reduced excitability in the sensorimotor cortex of LC individuals. Together, these features, which were associated with fatigue and brain fog, highlight altered dynamics in the sensorimotor cortex of LC individuals. These findings offer preliminary mechanistic insight into LC symptoms and suggest that PMBR and alpha CF warrant further investigation as candidate EEG markers of this condition.

1. Introduction

Long COVID (LC) affects 10–20% of individuals following SARS-CoV-2 infection, characterized by symptoms that persist beyond 12 weeks [1,3,105], including fatigue, cognitive dysfunction (“brain fog”), and post-exertional malaise (PEM) [24,64,67,103]. PEM, a hallmark of LC, involves symptom exacerbation after minor exertion [39]. Fatigue,

another core symptom, is characterized by persistent exhaustion that remains unrelieved by rest [83]. Cognitive dysfunction, reported by ~85% of LC individuals, impacts memory, attention, working memory, decision-making, and executive functioning [27]. Despite its high prevalence and disabling nature, LC lacks objective diagnostic tools and biomarkers [105].

Neurological involvement is increasingly recognized in COVID-19.

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SARS-CoV-2 presumably affects the central nervous system (CNS) through direct invasion across the blood–brain barrier (BBB) or via infected immune cells, with associated cytokine storms contributing to neuroinflammation and altered synaptic function [45,53,70,71,109]. Using magnetic resonance imaging (MRI), a cortical grey matter loss of ~2% was found post-SARS-CoV-2 infection, comparable to a decade of aging [32]. Reduced grey matter volume and white matter lesions observed post-infection were found to be associated with cognitive dysfunction, notably visuospatial impairment and global deficits across memory, language, and perceptual motor tasks [44,104]. For those with lingering symptoms evolving in the LC condition, thalamic volume loss and weakened thalamocortical connectivity with motor areas were reported, features that were associated with the level of fatigue [42,62]. In the brains of LC individuals, increased alpha power, slower alpha frequency, and elevated beta activity were also reported in region-specific areas using electroencephalography (EEG) at rest [21,60,74]. More specifically, reduced transcranial magnetic stimulation (TMS)-evoked beta activity was observed in motor-related regions, including the supplementary motor area (SMA), which was associated with fatigue severity [20]. Increased motor thresholds and reduced motor-evoked potentials were also observed in LC using TMS, indicating impaired corticospinal excitability, a feature associated with fatigue [79]. These findings confirm the presence of structural and functional brain changes post-SARS-CoV-2 infection and in the presence of LC, including reduced brain volume, impaired excitability, and altered alpha and beta oscillatory patterns [20,21,60,79,104].

In this study, we investigated the sensorimotor cortex (SMC), given its central role in movement-related beta dynamics and its modulation by attentional and sensorimotor demands. Specifically, alpha and beta oscillations in SMC were shown to vary during the orienting of attention to upcoming tactile stimuli, with spatially specific suppression of these rhythms linked to improved behavioral performance [98]. Additionally, beta-band activity in sensorimotor regions was shown to be modulated by attention and reflects higher-order processing of task-relevant sensory input, rather than motor execution alone [7]. In the SMC, beta power decreases during movement, a phenomenon known as motor-related beta desynchronization (MRBD) and rebounds afterward, referred to as post-movement beta rebound (PMBR) [19,82]. In older adults, MRBDs are enhanced, with greater values associated with degraded motor performance [107]. On the other hand, PMBR is reduced in elderly individuals, a phenomenon thought to reflect altered motor cortical inhibition and excitability [6,107]. As the brain seems to undergo accelerated age-related decline following SARS-CoV-2 infection [32,42], LC individuals may exhibit functional changes similar to those observed in older individuals.

We also examined alpha activity at rest, given its central role in regulating cortical excitability and supporting cognitive functions [12,57,58], as well as its reported changes in individuals with LC [21]. To capture these dynamics more precisely, resting EEG signals were decomposed into their periodic (oscillatory) and aperiodic (1/f background) components [31,96]. This approach was motivated by evidence suggesting that the aperiodic component is altered across neurodegenerative and psychiatric conditions [11,81,85,86], with some of these main features being linked to disease presence, stage and severity [81].

Reduced excitability in the SMC is a known characteristic of limbic-cortical dysfunction [20,79]. Central fatigue is believed to stem from diminished voluntary motor drive, which is associated with decreased excitability in motor-related cortical regions and impaired corticospinal transmission [23,34,54]. This implies that increasing cognitive–motor demands could further disrupt the brain's oscillatory activity in those affected by LC. To investigate this, participants learned a novel motor task, allowing the examination of whether post-task EEG dynamics would show greater deviations from those observed in healthy controls (HC).

Finally, whether these EEG features were associated with reported fatigue and cognitive impairment was examined, two primary symptoms

among LC individuals [46]. By linking resting and movement-related EEG features to fatigue and brain fog, this study sought to advance objective biomarkers for LC and estimate how altered cortical dynamics relate to its disabling symptoms.

Methods

1.1. Participants

This study, conducted at McGill University (May 2023–Sept. 2024), included LC participants who met the following inclusion criteria: (a) presence of fatigue, PEM, and brain fog, assessed using the Fatigue Severity Scale (FSS, score ≥ 36) [89], the DePaul Symptom Questionnaire for PEM (DSQ-PEM, frequency and severity score of 2 on items 1–5) [25], and Cognitive Failures Questionnaire (CFQ) [16]; (b) symptoms persisting for at least three months following a confirmed COVID-19 infection, verified by polymerase chain reaction (PCR) or antigen testing; (c) age between 18 and 59 years to reduce the influence of age-related brain changes; and (d) right-handedness, confirmed by the Edinburgh inventory of handedness [78]. Exclusion criteria included: (a) a history of neurological or psychiatric conditions, head injury, or epilepsy; (b) use of medications that potentially influence EEG activity, including CNS stimulants as well as anticonvulsants; and (c) a diagnosis of other conditions involving fatigue symptomatology. To ensure a clear comparison, HC were required to have no history of COVID-19 infection within the previous six months and no symptoms consistent with the WHO definition of LC [105].

The sample size was determined using an a priori power analysis conducted before recruitment to estimate the number of participants needed to detect baseline between-group differences in MRBDs (independent-samples t-test). Based on prior work [107], the expected effect size was large ($d \approx 1.12$; SDs 8.4 and 10.4). With a one-tailed α of 0.05 and 80% power, this corresponded to approximately 10–12 participants per group. A conservative minimum of 14 participants per group was therefore set.

Twenty-one participants were recruited from the Facebook page [Covid Longue Québec Long-Haulers Support Group](#) ($n = 2.9k$) and emailed through the *Quebec Action for Post-COVID Project*. Twenty-one HC were matched to the LC participants by age (± 3 years) and sex. Two participants (one from each group) were excluded due to excessive EEG noise, yielding a final analyzed sample of 20 LC and 20 control participants. The study was approved by the Research Ethics Office (IRB) of McGill University on March 27, 2023 (A02-M82-22B). All participants read and electronically signed the consent form via REDCap. Compensation was provided for transportation and participation.

1.2. Experimental paradigm

This was an exploratory cross-sectional study with matched pairs that included a repeated-measures component. The study consisted of two phases (Fig. 1). The first one involved an online assessment where participants completed eligibility and self-reported symptom questionnaires via REDCap platform. All materials were available in English and French. The second one involved an in-person laboratory session. At least a week after completing the online phase, EEG data were collected during rest and a handgrip task, both pre- and post-motor learning task (ML task). The ML task was used to increase the cognitive load. This task has been previously used in neurorehabilitation and skill learning studies, where brief, visually guided force tracking reliably engages neural systems without high exertion demands [26,75]. The second session lasted ~3 hours.

1.3. Data acquisition and pre-processing

1.3.1. Symptom questionnaires and behavioral assessment

In addition to the ones used for eligibility, questionnaires were used

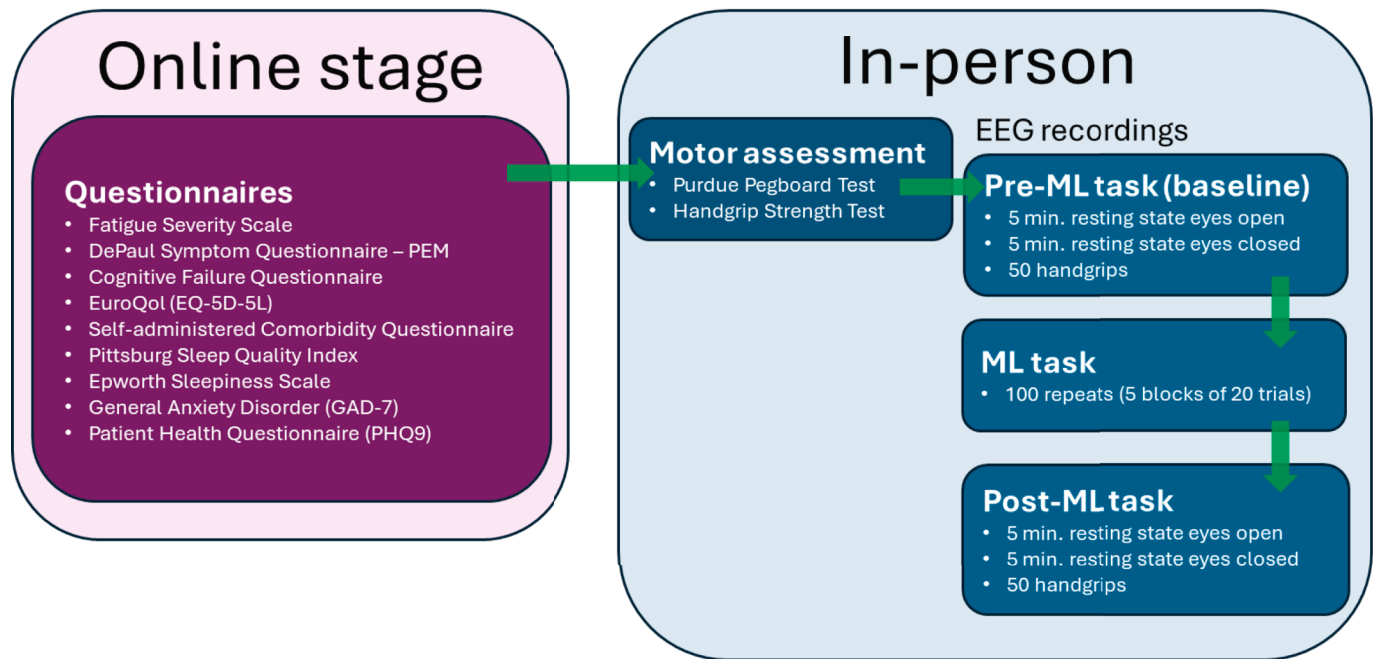


Fig. 1. Experimental paradigm flow chart. The study consists of two stages: an online stage and an in-person stage. During the online stage, participants complete a series of questionnaires assessing fatigue, cognitive failures, quality of life, comorbidities, sleep quality, and symptoms of anxiety and depression. The in-person stage includes a motor assessment with the Purdue Pegboard Test and Handgrip Strength Test, followed by EEG recordings during pre-ML (baseline), ML (motor learning), and post-ML task. EEG data collection includes resting-state recordings with eyes open and closed, as well as motor tasks involving handgrip repetitions.

to explore quality of life [41], daytime sleepiness [52], sleep quality [18], symptoms of anxiety [93] and depression [61], and comorbidities [87] (Fig.1). The Purdue Pegboard Test was used to assess hand and finger coordination and dexterity [65]. Separate scores were calculated for Left + Right + Both (L + R + B), as well as for the assembly task. Handgrip strength was used to determine the maximal voluntary contraction [14].

1.3.2. EEG data acquisition and pre-processing

EEG signals were recorded using a 64-channel high-density EEG (HD-EEG) system (Brain Products, Gilching, Germany), following the 10–20 electrode layout [56]. Signals were amplified and sampled at 2500 Hz. All electrodes were referenced to FCz, with AFz as ground. Electrode impedances were kept below 20 kΩ. Data were collected at rest for five minutes with eyes open, fixating on a cross in the center of the screen, followed by five minutes with eyes closed. Only eyes-open recordings were used in this study, as this state minimizes fluctuations in arousal and maintains stable visual engagement, providing a suitable baseline for analysis [9,10]. For the motor task, participants completed 50 right-hand contractions (4 seconds at 15% maximal voluntary contraction) using a dynamometer (BIOPAC, Goleta, CA, USA) with real-time visual feedback, following procedures routinely applied in our laboratory [72] (details provided in the [suppl. materials](#)). All participants were right-handed, confirmed via the Edinburgh Handedness Inventory [78], and the handgrips and ML task were performed with the right hand; C3 and C5 (contralateral, left-hemisphere electrodes) were therefore selected to capture task-relevant sensorimotor activity. Reaction time and rise time were extracted from the force signals. The ML task consisted of 100 trials of a force-matching task with real-time visual feedback and were performed as described in [26] (see [Supplementary Materials](#)).

EEG signals were filtered (0.1–120 Hz bandpass, 60 Hz notch), resampled at 250 Hz, and re-referenced to electrode FCz. Noisy epochs and channels (e.g., muscle, head, and jaw movement artifacts) were removed by visual inspection. Independent component analysis corrected eye movement artifacts (Brainstorm MATLAB toolbox) [95]. Frequency bands were set as: alpha (8–12 Hz) and beta (13–29 Hz).

Time-frequency representations were computed using Welch's method [102] with 10-second no-overlap windows. Power spectral density (PSD) estimates were obtained for each epoch and averaged within the predefined frequency bands (alpha, beta). Band power was extracted as the mean PSD within each frequency range for each time window of interest.

Although EEG was recorded using a 64-channel montage to ensure full scalp coverage and data quality, analyses were restricted a priori to electrodes C3 and C5 to test region-specific hypotheses related to SMC function. The same electrodes were used for resting alpha analyses to maintain anatomical consistency across resting-state and movement-related measures within the contralateral sensorimotor region. This choice was also supported by evidence that sensorimotor alpha and beta rhythms are modulated by somatosensory attention and behavioral performance [36,98].

Relative power (RP%), which represents the power changes relative to the baseline, was computed from electrodes C3 and C5, located over the left SMC, opposite to the right hand performing the task [29,84,90]. Brain imaging studies of EEG electrode localization show that C3/C4 consistently project onto the central sulcus region, spanning both pre-central (M1) and postcentral (S1) gyri, despite individual variability in the precise gyrus [59,77,90]. We therefore use the SMC consistent with prior studies targeting this same region under the same terminology [72,108]. Time windows were defined as: Baseline (pre-movement): 1.0 - 0.1 seconds preceding movement onset (see [Figs. 2 & 3](#)). MRBD (active contraction): 0.5 - 3.5 seconds post-movement onset. PMBR (rebound phase) 5 - 8 seconds post-movement onset. These windows were selected based on established temporal profiles reported in prior work [82,107], with visual inspection used as a confirmatory step to verify alignment with the observed signal patterns. RP% was calculated using the following formula, where $P(t)$ is the absolute power at time t , and B is the baseline power:

$$RP\% = \frac{P(t) - B}{B} \times 100\%$$

Periodic and aperiodic decomposition using the specparam algorithm

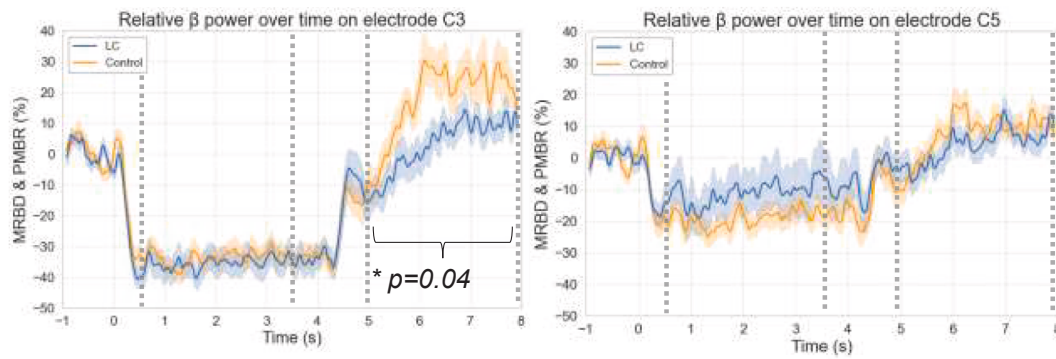


Fig. 2. Temporal evolution of the beta power relative to the baseline (MRBD & PMBR) in electrodes C3 and C5 during the gripping task. Handgrips were held from 0 to 4 s. The Long COVID (LC) group is shown in blue and the control group in orange. The gray dotted lines delimit the time intervals used to quantify beta power changes. MRBD was derived from the movement-related window (0.5–3.5 s), and PMBR from the post-movement rebound window (5–8 s). * $p < 0.05$ indicates significant differences in the power at a specific time.

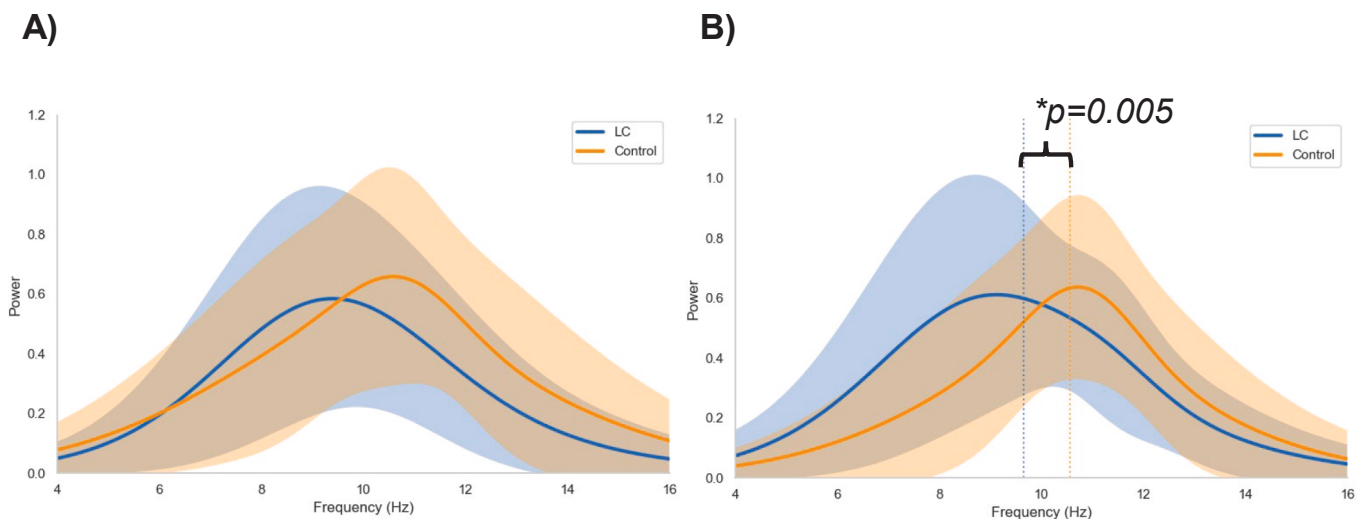


Fig. 3. Alpha-band power spectra during eyes-open resting-state at electrodes C3 and C5. A) Power spectral density at electrode C3; B) Power spectral density at electrode C5. The lines indicate the group averages and the shadow represents the standard deviation (SD). The control group is shown in orange, and the long COVID (LC) group in blue. A significant group difference in alpha central frequency (CF) is observed at electrode C5 ($p = 0.005$).

(formerly Fitting Oscillations and One-Over-F (FOOOF)) [31,38,80]) was conducted on the resting-state recordings from electrodes C3 and C5. Resting-state recordings were segmented into 10-second windows, which were averaged to reduce the impact of transient artifacts and improve spectral estimation reliability [102]. The aperiodic exponent and offset were extracted from the aperiodic and the alpha central frequency (CF; equivalent to peak alpha frequency), peak power (PW), and bandwidth (BW) of the periodic alpha component. The algorithm was constrained to identify a single peak, ensuring that extracted parameters reflected the dominant alpha component and allowing consistent comparisons across participants.

1.4. Data analysis

1.4.1. Symptoms, cognitive and behavioral data analysis

Group comparisons were conducted using independent-sample t-tests or the Mann-Whitney U Test for non-normal data. ML task performance across blocks was analyzed using a linear mixed model (LMM) to assess group differences and changes over time.

1.4.2. EEG data analysis

Normality was assessed using the Shapiro-Wilk test. Group comparisons of MRBD, PMBR, periodic and aperiodic components were

conducted using independent-sample t-tests or Mann-Whitney U tests for non-normally distributed data. LMM assessed changes from pre- to post-task.

Correlations between EEG features and clinical symptoms were calculated using Pearson or Kendall's tau-b correlation. Bonferroni correction was applied to hypothesis-driven tests, and FDR correction was applied to exploratory analyses (reported in the [Supplementary Material](#)). Group and LC-only correlations were both reported to capture full and within-group associations.

2. Results

2.1. Self-reported clinical symptoms and behavioral results

Out of the 42 participants recruited, two (one from each group) were excluded due to excessive noise in their EEG recordings, leaving 20 participants per group. For the LC group, the average time between the confirmed COVID-19 infection and the recruitment date was 24.0 ± 10.7 months. Pre-task EEG recordings were successfully obtained from all 20 participants in both groups. In the LC group, three participants had incomplete post-task recordings due to discomfort with the EEG cap and system malfunctions.

The groups did not differ in age and sex ([Table 1](#)). The scores from all

Table 1

Demographic group variables and results from the scores of the questionnaires.

	LC	control		
	mean (SD)	mean (SD)	<i>p</i>	Test Statistic
Age	42.7 (9.7)	42.5 (10.2)	0.66	<i>t</i> (37.9) = 0.06
Sex (female)	15	15	–	–
COVID-19 hospitalizations	1	0	–	–
Time since COVID infection (months)	24 (10.7)	N/A	–	–
Fatigue Severity Scale (FSS) ¹	56 (5.6)	23.5 (12.2)	<.001 ^{1*}	U = 9.50
Cognitive Failures Questionnaire (CFQ)	58.5 (14.1)	27 (11.6)	0<.001 ^{1*}	<i>t</i> (38) = 7.68
DePaul Symptom Questionnaire-PEM (DSQ-PEM)	100%	0%	–	–
Epworth Sleepiness Scale (ESS)	10.8 (5.0)	4.6 (2.3)	0<.001 ^{1*}	<i>t</i> (38) = 7.68
General Anxiety Disorder (GAD7) ¹	5.9 (4.3)	3.5 (5.1)	0.02 ^{1*}	U = 114.00
Patient Health Questionnaire (PHQ-9) ¹	10.2 (4.8)	2.5 (3.0)	<.001 ^{1*}	U = 29.00
Self-Administered Comorbidities Questionnaire ¹	4.6 (4.3)	0.5 (1.1)	<.001 ^{1*}	U = 75.00
Pittsburgh Sleep Quality Index (PSQI)	10.9 (3.0)	5.1 (2.7)	0<.001 ^{1*}	<i>t</i> (38) = 6.70
EuroQol EQ-5D-5 L ¹	41.5 (18.0)	86.6 (6.4)	0.001 ^{1*}	U = 0.00

¹For non-parametric tests, the Mann-Whitney *U* test was used, and the *U* value is reported instead of *t*(*df*). **p* < 0.05 indicates statistically significant differences.

the questionnaires are summarized in Table 1. All LC individuals met our inclusion criteria. They experienced higher levels of fatigue and brain fog compared to HC (FSS & CFQ: *p* < 0.001). The scores from all other questionnaires, used to quantify sleep quality, daytime sleepiness, quality of life, comorbidities, and symptoms of anxiety and depression, differed between groups (*p* < 0.05). The Purdue Pegboard Test scores were lower in the LC group for the L + R + B (*p* < 0.01), but not for the assembly scores (Table 2). No differences were observed in the hand grip strength scores between groups for both hands.

For parametric tests, the *t*-statistic is reported with degrees of freedom in parentheses (*t*(*df*)).

Table 2

Results from the scores of the Purdue Pegboard Test (PPT), Handgrip Strength Test, and Reaction and Rise Times extracted from the dynamometer.

	LC	Control		
	Mean (SD)	Mean (SD)	<i>p</i>	Test Statistic
PPT				
L + R + B ¹	38.9 (7.4)	43.1 (3.9)	0.03 ^{1*}	U = 279.5
Assembly	32.4 (8.6)	35.5 (6.7)	0.20	<i>t</i> (38) = 1.2
Handgrip Strength Test				
Right	30.9 (8.4)	30.5 (7.9)	0.88	<i>t</i> (38) = −0.1
Left	28.4 (7.8)	28.6 (7.9)	0.94	<i>t</i> (38) = 0.07
Reaction time				
Pre-task	0.75 (0.1)	0.61 (0.1)	0.02 ^{1*}	<i>t</i> (37) = −2.3
Post-task	0.75 (0.08)	0.67 (0.3)		
Rise time				
Pre-task	0.29 (0.08)	0.34 (0.07)	0.12	<i>t</i> (37) = 1.5
Post-task	0.32 (0.08)	0.33 (0.05)		

For parametric tests, the *t*-statistic is reported with degrees of freedom in parentheses (*t*(*df*)).

¹For non-parametric tests, the Mann-Whitney *U* test was used, and the *U* value is reported instead of *t*(*df*). **p* < 0.05 indicates statistically significant differences.

The LC group had slower reaction times than the HC (*p* = 0.02), while baseline rise time did not differ between groups. The pre- to post-task scores did not differ between groups in terms of reaction (*p* = 0.52) and rise times (*p* = 0.07) during the handgrip task.

For the ML task, there was a significant effect of time on motor learning scores (*F*(4, 150.92) = 54.22, *p* < 0.001), indicating an increase in the scores across trials. However, there was no significant main effect of group (*F*(1, 42.709) = 1.90, *p* = 0.17), nor a significant group × time interaction (*F*(4, 150.92) = 1.057, *p* = 0.38) (see [Suppl. Material](#)).

2.2. EEG features during movement

Grand averages of the temporal evolution of the beta band power relative to the baseline during the gripping task are shown in Fig. 2. No differences were found between groups in MRBD values in electrode C3 (*U* = 193.00, *p* = 0.86; HC: *M* = −33.64, *SD* = 13.13; LC: *M* = −35.04, *SD* = 17.29) or electrode C5 (*U* = 161.00, *p* = 0.29; HC: *M* = −18.97, *SD* = 13.64; LC: *M* = −12.61, *SD* = 25.50).

For the PMBR values, a significant difference was observed at electrode C3 (*t*(38) = 2.36, *p* = 0.02), where the mean for the HC group (*M* = 17.47, *SD* = 21.01) was higher than that of the LC group (*M* = 2.95, *S* = 17.63). No differences were observed at electrode C5 (*t*(38) = −1.38, *p* = 0.17; HC: *M* = 8.04, *SD* = 11.63; LC: *M* = 2.92, *SD* = 11.75).

There were no differences in the effect of group × time (pre-task vs. post-task) for either MRBDs or PMBRs. This is shown in the [Supplementary Material](#).

2.3. EEG power spectral during resting state

Differences between groups were found in the alpha CF at electrode C3 (*p* = 0.03), where the control group had a higher value (10.5 ± 0.83) than the LC group (9.8 ± 1.16), *t*(35) = −2.27. A similar trend was observed at electrode C5 (*p* = 0.005), with the HC group having a higher power (10.6 ± 0.6) than the LC group (9.6 ± 1.0), *t*(34) = −3.06 (Fig. 3). However, after adjusting for multiple comparisons, only the results from electrode C5 remained significant.

No differences were found in the alpha PW on electrode C3 (*t*(35) = 0.21, *p* = 0.83) or C5 (*t*(34) = −0.43, *p* = 0.66) nor in the alpha BW on electrode C3 (*U* = 196.00, *p* = 0.45) or C5 (*U* = 145.00, *p* = 0.64).

No differences were found in the aperiodic components such as the exponent in electrode C5 (*t*(38) = −0.74, *p* = 0.46) or C3 (*t*(38) = 0.01, *p* = 0.98), nor in the offset in electrode C5 (*U* = 140.50, *p* = 0.11) or C3 (*t*(38) = −0.60, *p* = 0.54).

No differences were found in the effect of group × time (pre-task vs. post-task) for any of the periodic and aperiodic components at electrode C3 or C5. This is shown in the [Supplementary Material](#).

2.4. Association between EEG components, questionnaires, and motor scores

A negative correlation was observed between CFQ scores and PMBR at electrode C3 (*r* = −0.35, *p* = 0.02) (Fig. 4A). No correlation was found between the PMBR at electrode C3 and the FSS questionnaire scores (*r* = −0.13, *p* = 0.22) (Fig. 4B).

The alpha CF at electrode C5 was negatively correlated with both the CFQ (*r* = −0.23, *p* = 0.04) (Fig. 4C) and the FSS scores (*r* = −0.26, *p* = 0.01) (Fig. 4D), although only the correlation with FSS remained significant after correction for multiple comparisons. Neither PMBR at C3 nor alpha CF at C5 correlated significantly with time since infection (*r* = −0.12, *p* = 0.62; *r* = −0.33, *p* = 0.17, respectively), suggesting that the observed group differences were not simply a function of variability in illness duration within the sample.

No other significant correlations were found in the additional exploratory analyses between motor performance and EEG components, whether analyzing the full sample or only the LC group (see [Suppl. Materials](#)).

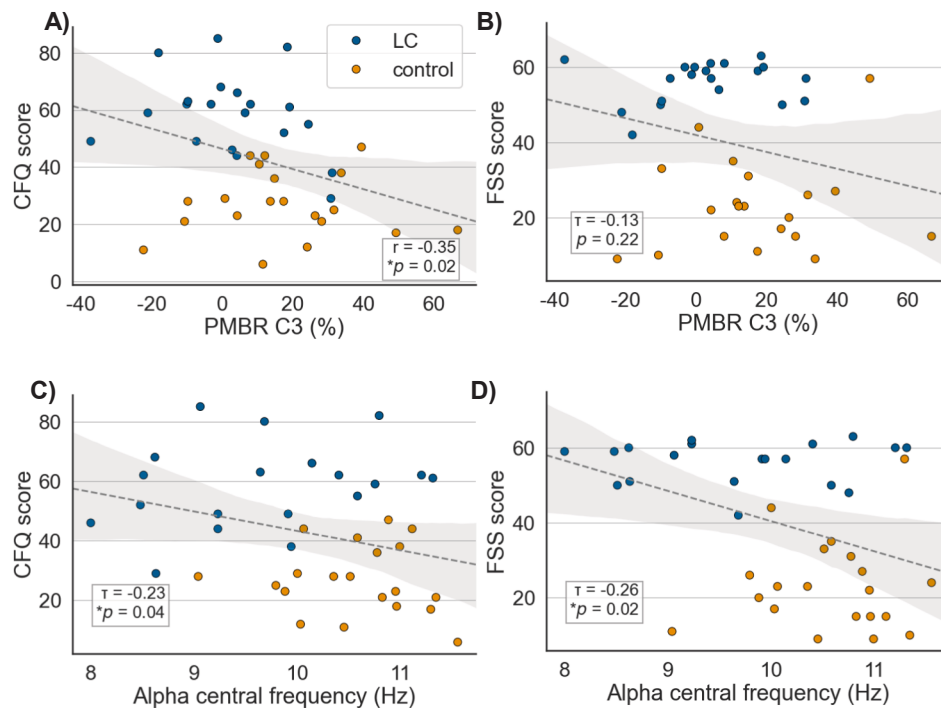


Fig. 4. Relationship between the localized EEG features and self-reported clinical symptom scores. A) Post-movement beta rebound (PMBR) at electrode C3 and cognitive failures questionnaire (CFQ) scores. B) PMBR at electrode C3 and fatigue severity scale (FSS) scores. C) Alpha central frequency (CF) at electrode C5 and CFQ scores. D) Relationship between alpha CF at electrode C5 and FSS score. For each graph, correlation coefficients and p-values are displayed. Significant correlations ($p < 0.05$) are marked with an asterisk (*).

3. Conclusions

To our knowledge, this is the first study to assess how EEG oscillatory patterns are modulated in the SMC of LC individuals during movement, as well as their periodic and aperiodic characteristics. LC participants showed reduced beta activity post-movement, which was associated with the severity of self-reported cognitive symptoms. During rest, they showed a lower alpha CF, which was linked to fatigue severity. These findings show that cortical oscillations are disrupted in LC symptoms, and they might contribute to the symptomatology observed in this population.

3.1. Self-reported clinical symptoms and behavioral results

LC participants exhibited severe fatigue, brain fog, and PEM, consistent with previous studies [16,27,28,49,89,97,110]. They also reported increased symptoms of anxiety, symptoms of depression, poor sleep quality, daytime sleepiness, comorbidities, and reduced quality of life, features observed consistently in this population [27,30,68,76,91,94].

Motor performance was affected in the LC group, where poorer bimanual coordination and fine motor control were present [65]. No differences were found in the assembly task or grip strength, suggesting coordination may be more affected than strength or precision. Slower reaction times were observed during the handgrip task in LC participants, while rise time did not differ between groups. Reaction time is broadly recognized as a sensitive marker of neurobiological integrity and processing efficiency [40]. This pattern of delayed initiation with preserved motor output is consistent with prior reports of slower reaction times in LC, observed across visual detection tasks and neuropsychological assessments [22,69].

3.2. EEG-based MRBD & PMBR

PMBRs in the SMC were reduced in LC individuals. This feature is believed to reflect motor cortical inhibition following movement [55] and to be mediated by GABAergic neurotransmission [33,37]. Reduced PMBR has been reported in Parkinson's disease and multiple sclerosis, linked to impaired inhibitory control, diminished motor accuracy, and fatigue in these conditions [6,63,66]. Reduced beta activity post-movement in LC suggests disrupted GABAergic function. This aligns with observations of reduced TMS-evoked beta responses in individuals with LC who experience high levels of fatigue, where higher fatigue correlated with reduced beta activity in the SMC [20].

Beta activity in the SMC is thought to arise from corticothalamic and cortico-basal ganglia loops [8,15]. In aging, reduced PMBR has been linked to impaired thalamocortical connectivity [47]. Comparable findings in LC, including structural disruptions in the thalamus and basal ganglia [19], and weakened thalamocortical coupling with M1 and SMA in fatigued individuals [20], suggest that altered thalamocortical circuits may underlie impaired regulation of motor cortical excitability.

3.3. EEG power spectral analysis

LC participants showed lower alpha band CF at rest over the SMC. Reduced alpha peak frequency has been reported in aging and several neurological conditions associated with cognitive impairment, although these findings have generally been derived from global, posterior, or non-sensorimotor electrode sets rather than specifically from C3/C5 [21,35,111]. We prioritized anatomical consistency between our resting-state and movement-related measures, both drawn from C3/C5, since our primary aim was to link rest and movement-related activity within the same sensorimotor region. This choice limits direct comparability with prior alpha CF findings. Alpha CF is thought to reflect the temporal organization of cortical processing, with slower frequencies associated with less efficient information integration and reduced

cognitive speed. Accordingly, a lower alpha CF may indicate a shift in the balance of excitation and inhibition, potentially disrupting neural communication and cognitive processing [50,88]. Cognitive efficiency can be affected by chronic systemic inflammation, which has been shown to alter synaptic function and large scale network dynamics, leading to disruptions in alpha oscillatory activity and reduced cortical network stability, a process also implicated in LC [50,101].

Alpha oscillations are thought to depend on thalamocortical circuits that regulate cortical activity [48]. In LC, both slower alpha CF and reduced PMBR findings point to disruptions in the thalamocortical and basal ganglia circuits. Evidence of structural abnormalities and weakened thalamocortical coupling in fatigued LC individuals further supports this interpretation [42,62].

There were no differences in the periodic and aperiodic measures extracted from the alpha band (PW, BW, exponent, offset). Previous EEG studies in LC reported increased alpha power but did not separate periodic and aperiodic components, leaving it unclear whether the observed effects reflected genuine oscillatory changes or broadband power shifts [21,60,74]. By applying this method, our approach provides a more specific characterization, indicating that the alterations in LC are more likely related to synchronized oscillatory dynamics rather than global cortical excitability [31].

No differences in spectral components were observed post-task, suggesting that these features may be relatively resistant to short-term modulation by cognitive load. Alternatively, the small sample size and inter-individual variability may have reduced sensitivity to detect subtle changes.

3.4. Associations between EEG variables, self-reported symptoms and behavioral scores

A negative correlation was found between brain fog, measured through cognitive failures, and beta activity in the SMC after movement. This indicates that greater cognitive difficulties were associated with reduced PMBR. PMBR is thought to index inhibitory control and the resetting of motor networks [17,55], processes that depend on GABAergic tone [33,73]. In LC, a reduced PMBR may therefore reflect GABAergic dysfunction contributing to both brain fog and fatigue [20,79]. Given the SMC's involvement in motor and cognitive functions, reduced PMBR could therefore reflect abnormalities in a shared neural substrate underlying these symptoms [51,100].

A lower alpha CF was associated with greater fatigue in our sample. This finding is consistent with prior post-COVID EEG work showing alpha frequency slowing, although the electrode locations and symptom domains differed from those examined here [21]. Therefore, these results are not directly comparable to the present C3/C5 EEG-based measure. Alpha band slowing is observed in mild cognitive impairment, multiple sclerosis, burnout, and Alzheimer's disease, where it is associated with deficits in executive function, working memory, attention, and mental fatigue [5,50,99]. Reduced PMBR has also been reported in Parkinson's disease [43], multiple sclerosis [2], amyotrophic lateral sclerosis [13], and major depressive disorder [106]. This suggests a transdiagnostic marker of impaired sensorimotor inhibition rather than an LC-specific one. A recent systematic review similarly found overlapping alpha- and beta-band abnormalities across fibromyalgia, chronic fatigue syndrome (ME/CFS), and LC [92], further indicating that these features likely reflect a shared response to chronic fatigue-related states. These findings suggest reduced alpha CF could serve as a neurophysiological marker for LC-related cognitive and fatigue symptoms. This interpretation is further in line with prior work linking disruptions in alpha rhythms to brain fog in LC [4,60].

Within the LC group alone, neither PMBR nor alpha CF correlated significantly with fatigue or cognitive symptoms (see [Suppl. Table 1](#)), suggesting that associations observed in the combined sample partly reflect between-group differences in symptom severity rather than a continuous symptom-EEG relationship within LC.

A trend-level association was observed between cognitive failures and alpha CF, suggesting that lower alpha frequencies may be associated with cognitive symptoms. Although not statistically significant, this aligns with previous reports [5,99] and warrants further investigation.

Exploratory correlations (see [suppl. materials](#)) revealed several emerging trends. In the full sample, MRBDs were associated with higher depression symptoms and cognitive failures, while smaller PMBRs were linked to worse sleep quality, increased cognitive failures, symptoms of depression, and increased comorbidities. In the LC group, MRBDs correlated positively with daytime sleepiness and negatively with the number of comorbidities, whereas PMBR was positively linked with sleep quality and comorbidities. Depression symptoms were negatively associated with alpha CF and positively associated with alpha PW, while alpha BW was positively correlated with anxiety symptoms. These findings provide direction for future hypothesis-driven studies with larger LC samples.

As an exploratory study, a primary limitation is the small sample size. This reduces sensitivity to detect subtle group differences or associations between EEG markers and clinical symptoms. Given the sample size, the study was underpowered to detect small-to-moderate effects in secondary and exploratory analyses, and the absence of significant findings in these analyses should not be interpreted as evidence of no effect. For instance, LC participants performed below controls across learning blocks during the ML task. Although this difference was not statistically significant, the pattern may reflect subtle impairments in adaptive motor processes potentially influenced by fatigue and cognitive symptoms. Task-related effects should also be considered when interpreting PMBR, as even low-effort tasks may induce mental or physical fatigue in LC participants, making it difficult to disentangle task-induced strain from underlying condition-related fatigue. These consistent trends, together with the sample size limitation, suggest that subtle group effects may have been masked by variability and could become more evident in larger or longitudinal cohorts. Correlation analyses between EEG features and clinical symptoms should be interpreted with caution. Associations were examined across the full sample and may be driven by group differences in symptom severity between LC and controls. Within-group analyses did not consistently reach statistical significance, limiting conclusions about symptom-EEG relationships within LC. As this was a cross-sectional study, it cannot determine whether the observed EEG alterations change over the course of illness. Because analyses were a priori restricted to electrodes over the left SMC, our findings cannot be generalized to broader cortical dysfunction or other brain regions. Additionally, as all participants were right-handed, our findings are specific to the dominant left SMC and may not generalize to left-handed or ambidextrous individuals. We also did not control for potential confounders that may impact both symptom presentation and brain activity, such as hormonal variation, age, and depressive symptoms. Furthermore, LC participants differed from controls in sleep quality, daytime sleepiness, anxiety, depression, quality of life, and comorbidity burden, none of which were statistically controlled for in the present analyses. In addition, participants varied considerably in time since infection and longitudinal studies are needed to determine whether cortical oscillatory changes evolve over the course of LC. As several of these variables are known to influence alpha and beta oscillatory activity, the present findings should be interpreted as reflecting the LC phenotype broadly. PEM was monitored given its relevance as a hallmark symptom of LC; however, it was assessed using a binary classification (DSQ-PEM), limiting the ability to examine associations with EEG features. In addition, the study was not designed to provoke PEM, making it difficult to capture its relationship with neural activity or to distinguish it from general fatigue. Future work should extend this paradigm to test whether these EEG markers generalize across different patterns of hemispheric dominance.

In conclusion, this study provides evidence of altered neural oscillatory dynamics in LC, with a slower alpha frequency at rest and reduced beta rebound following movement. These findings highlight disrupted

inhibitory processes and an imbalance between excitation and inhibition, likely involving thalamocortical and basal ganglia circuits. Importantly, both markers were associated with the severity of fatigue and cognitive dysfunction, linking altered brain activity to the core symptoms of LC. By identifying candidate EEG features associated with persistent symptoms, this exploratory work represents a step toward the future development of objective biomarkers for LC.

Ethics statement

The study was conducted in accordance with the Declaration of Helsinki and approved by the McGill University Ethical Advisory Committee #A02-M82-22B on March 27, 2023.

Informed consent statement

Informed consent was obtained from all participants involved in the study.

CRedit authorship contribution statement

M. Casado Sánchez: Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **K. Morales Fajardo:** . **S. Long:** Visualization, Conceptualization. **T. Janaudis-Ferreira:** Writing – review & editing, Supervision. **C. Duclos:** Writing – review & editing, Supervision. **M.H. Boudrias:** Writing – review & editing, Validation, Supervision, Resources, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Ethics declaration

Studies in human

This study was performed in compliance with relevant laws, regulatory frameworks and guidelines where the research took place.

Written informed consent to take part in the study and to publish the article has been obtained from all participants or their legal representatives. The privacy rights of participants have been observed.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jocn.2026.112230>.

References

- [1] Al-Aly Z, Bowe B, Xie Y. Long COVID after breakthrough SARS-CoV-2 infection. *Nat Med* 2022;28(7):1461–7. <https://doi.org/10.1038/s41591-022-01840-0>.
- [2] Arpin DJ, Heinrichs-Graham E, Gehringer JE, Zabab R, Wilson TW, Kurz MJ. Altered sensorimotor cortical oscillations in individuals with multiple sclerosis suggests a faulty internal model. *Hum Brain Mapp* 2017;38(8):4009–18. <https://doi.org/10.1002/hbm.23644>.
- [3] Ayoubkhani D, Bosworth ML, King S, Pouwels KB, Glickman M, Nafilyan V, et al. Risk of long COVID in people infected with severe acute respiratory syndrome coronavirus 2 After 2 doses of a coronavirus disease 2019 vaccine: community-based, matched cohort study. *Open Forum Infect Dis* 2022;9(9):ofac464. <https://doi.org/10.1093/ofid/ofac464>.
- [4] Babiloni C, Gentilini Cacciola E, Tucci F, Vassalini P, Chilovi A, Jakhar D, et al. Resting-state EEG rhythms are abnormal in post COVID-19 patients with brain fog without cognitive and affective disorders. *Clin Neurophysiol* 2024;161:159–72. <https://doi.org/10.1016/j.clinph.2024.02.034>.
- [5] Babiloni C, Jakhar D, Tucci F, Del Percio C, Lopez S, Soricelli A, et al. Resting state electroencephalographic alpha rhythms are sensitive to Alzheimer's disease mild cognitive impairment progression at a 6-month follow-up. *Neurobiol Aging* 2024;137:19–37. <https://doi.org/10.1016/j.neurobiolaging.2024.01.013>.
- [6] Bardouille T, Bailey L. Evidence for age-related changes in sensorimotor neuromagnetic responses during cued button pressing in a large open-access dataset. *Neuroimage* 2019;193:25–34. <https://doi.org/10.1016/j.neuroimage.2019.02.065>.
- [7] Bardouille T, Picton TW, Ross B. Attention modulates beta oscillations during prolonged tactile stimulation. *Eur J Neurosci* 2010;31(4):761–9. <https://doi.org/10.1111/j.1460-9568.2010.07094.x>.
- [8] Barone J, Rossiter HE. Understanding the role of sensorimotor beta oscillations. *Front Syst Neurosci* 2021;15:655886. <https://doi.org/10.3389/fnsys.2021.655886>.
- [9] Barry RJ, Clarke AR, Johnstone SJ, Magee CA, Rushby JA. EEG differences between eyes-closed and eyes-open resting conditions. *Clin Neurophysiol* 2007;118(12):2765–73. <https://doi.org/10.1016/j.clinph.2007.07.028>.
- [10] Barry RJ, De Blasio FM. EEG differences between eyes-closed and eyes-open resting remain in healthy ageing. *Biol Psychol* 2017;129:293–304. <https://doi.org/10.1016/j.biopsycho.2017.09.010>.
- [11] Belova EM, Semenova U, Gamaleya AA, Tomskiy AA, Sedov A. Voluntary movements cause beta oscillations increase and broadband slope decrease in the subthalamic nucleus of parkinsonian patients. *Eur J Neurosci* 2021;53(7):2205–13.
- [12] Berger, H. (1929). Über das elektroenkephalogramm des menschen. *Archiv für psychiatrie und nervenkrankheiten*, 87(1), 527–570.
- [13] Bizovičar N, Drejo J, Koritnik B, Zidar J. Decreased movement-related beta desynchronization and impaired post-movement beta rebound in amyotrophic lateral sclerosis. *Clin Neurophysiol* 2014;125(8):1689–99. <https://doi.org/10.1016/j.clinph.2013.12.108>.
- [14] Bohannon RW, Peolsson A, Massy-Westropp N, Desrosiers J, Bear-Lehman J. Reference values for adult grip strength measured with a Jamar dynamometer: a descriptive meta-analysis. *Physiotherapy* 2006;92(1):11–5. <https://doi.org/10.1016/j.physio.2005.05.003>.
- [15] Brittain JS, Brown P. Oscillations and the basal ganglia: motor control and beyond. *Neuroimage* 2014;85(Pt 2):637–47. <https://doi.org/10.1016/j.neuroimage.2013.05.084>.
- [16] Broadbent DE, Cooper PF, FitzGerald P, Parkes KR. The cognitive failures questionnaire (CFQ) and its correlates. *Br J Clin Psychol* 1982;21(1):1–16. <https://doi.org/10.1111/j.2044-8260.1982.tb01421.x>.
- [17] Brown P. Abnormal oscillatory synchronisation in the motor system leads to impaired movement. *Curr Opin Neurobiol* 2007;17(6):656–64. <https://doi.org/10.1016/j.conb.2007.12.001>.
- [18] Buysse DJ, Reynolds 3rd CF, Monk TH, Berman SR, Kupfer DJ. The pittsburgh sleep quality index: a new instrument for psychiatric practice and research. *Psychiatry Res* 1989;28(2):193–213. [https://doi.org/10.1016/0165-1781\(89\)90047-4](https://doi.org/10.1016/0165-1781(89)90047-4).
- [19] Cassim F, Szurhaj W, Sediri H, Devos D, Bourriez J-L, Poirot I, et al. Brief and sustained movements: differences in event-related (de)synchronization (ERD/ERS) patterns. *Clin Neurophysiol* 2000;111(11):2032–9. [https://doi.org/10.1016/S1388-2457\(00\)00455-7](https://doi.org/10.1016/S1388-2457(00)00455-7).
- [20] Casula EP, Esposito R, Dezi S, Ortellì P, Sebastianelli L, Ferrazzoli D, et al. Reduced TMS-evoked EEG oscillatory activity in cortical motor regions in patients with post-COVID fatigue. *Clin Neurophysiol* 2024;165:26–35. <https://doi.org/10.1016/j.clinph.2024.06.008>.
- [21] Cecchetti G, Agosta F, Canu E, Basaia S, Barbieri A, Cardamone R, et al. Cognitive, EEG, and MRI features of COVID-19 survivors: a 10-month study. *J Neurol* 2022;269(7):3400–12. <https://doi.org/10.1007/s00415-022-11047-5>.

- [22] Charles James J, Schulze H, Siems N, Prehn C, Quast DR, Trampe N, et al. Neurological post-COVID syndrome is associated with substantial impairment of verbal short-term and working memory. *Sci Rep* 2025;15(1):1695. <https://doi.org/10.1038/s41598-025-85919-x>.
- [23] Chaudhuri A, Behan PO. Fatigue and basal ganglia. *J Neurol Sci* 2000;179(S 1–2):34–42. [https://doi.org/10.1016/S0022-510X\(00\)00411-1](https://doi.org/10.1016/S0022-510X(00)00411-1).
- [24] Chen AK, Wang X, McCluskey LP, Morgan JC, Switzer JA, Mehta R, et al. Neuropsychiatric sequelae of long COVID-19: Pilot results from the COVID-19 neurological and molecular prospective cohort study in Georgia, USA. *Brain Behav Immun Health* 2022;24:100491. <https://doi.org/10.1016/j.bbih.2022.100491>.
- [25] Cotler J, Holtzman C, Dudun C, Jason LA. A brief questionnaire to assess post-exertional malaise. *Diagnostics (Basel)* 2018;8(3). <https://doi.org/10.3390/diagnostics8030066>.
- [26] Dal Maso F, Desormeau B, Boudrias M-H, Roig M. Acute cardiovascular exercise promotes functional changes in cortico-motor networks during the early stages of motor memory consolidation. *Neuroimage* 2018;174:380–92. <https://doi.org/10.1016/j.neuroimage.2018.03.029>.
- [27] Davis HE, Assaf GS, McCorkell L, Wei H, Low RJ, Re'em Y, et al. Characterizing long COVID in an international cohort: 7 months of symptoms and their impact. *eClinicalMedicine* 2021;38. <https://doi.org/10.1016/j.eclinm.2021.101019>.
- [28] Davis HE, McCorkell L, Vogel JM, Topol EJ. Long COVID: major findings, mechanisms and recommendations. *Nat Rev Microbiol* 2023;21(3):133–46. <https://doi.org/10.1038/s41579-022-00846-2>.
- [29] de Klerk CJM, Johnson MH, Southgate V. An EEG study on the somatotopic organisation of sensorimotor cortex activation during action execution and observation in infancy. *Dev Cogn Neurosci* 2015;15:1–10. <https://doi.org/10.1016/j.dcn.2015.08.004>.
- [30] Diem L, Schwarzwald A, Friedli C, Hammer H, Gomes-Fregolente L, Warncke J, et al. Multidimensional phenotyping of the post-COVID-19 syndrome: a Swiss survey study. *CNS Neurosci Ther* 2022;28(12):1953–63. <https://doi.org/10.1111/cns.13938>.
- [31] Donoghue T, Haller M, Peterson EJ, Varma P, Sebastian P, Gao R, et al. Parameterizing neural power spectra into periodic and aperiodic components. *Nat Neurosci* 2020;23(12):1655–65. <https://doi.org/10.1038/s41593-020-00744-x>.
- [32] Douaud G, Lee S, Alfaro-Almagro F, Arthofer C, Wang C, McCarthy P, et al. SARS-CoV-2 is associated with changes in brain structure in UK Biobank. *Nature* 2022;604(7907):697–707. <https://doi.org/10.1038/s41586-022-04569-5>.
- [33] Gaetz W, Edgar JC, Wang DJ, Roberts TP. Relating MEG measured motor cortical oscillations to resting γ -aminobutyric acid (GABA) concentration. *Neuroimage* 2011;55(2):616–21. <https://doi.org/10.1016/j.neuroimage.2010.12.077>.
- [34] Gandevia SC, Allen GM, McKenzie DK. Central fatigue. Critical issues, quantification and practical implications. *Adv Exp Med Biol* 1995;384:281–94.
- [35] Grandy TH, Werkle-Bergner M, Chicherio C, Schmiedek F, Lövdén M, Lindenberger U. Peak individual alpha frequency qualifies as a stable neurophysiological trait marker in healthy younger and older adults. *Psychophysiology* 2013;50(6):570–82. <https://doi.org/10.1111/psyp.12043>.
- [36] Haegens S, Händel BF, Jensen O. Top-down controlled alpha band activity in somatosensory areas determines behavioral performance in a discrimination task. *J Neurosci* 2011;31(14):5197–204. <https://doi.org/10.1523/JNEUROSCI.5199-10.2011>.
- [37] Hall SD, Stanford IM, Yamawaki N, McAllister CJ, Rönqvist KC, Woodhall GL, et al. The role of GABAergic modulation in motor function related neuronal network activity. *Neuroimage* 2011;56(3):1506–10. <https://doi.org/10.1016/j.neuroimage.2011.02.025>.
- [38] Haller M, Donoghue T, Peterson E, Varma P, Sebastian P, Gao R, Noto T, Knight R, T., Shestuyuk A., & Voytek, B. (2018). Parameterizing neural power spectra. *bioRxiv*, 299859. <https://doi.org/10.1101/299859>.
- [39] Hartle M, Bateman L, Vernon SD. Dissecting the nature of post-exertional malaise. *Fatigue: Biomed, Health Behav* 2021;9(1):33–44. <https://doi.org/10.1080/21641846.2021.1905415>.
- [40] Haynes BJ, Bauermeister S, Bunce D. A systematic review of longitudinal associations between reaction time intraindividual variability and age-related cognitive decline or impairment, dementia, and mortality. *J Int Neuropsychol Soc* 2017;23(5):431–45. <https://doi.org/10.1017/s1355617717000236>.
- [41] Hegde S, Sreeram S, Bhat KR, Satish V, Shekar S, Babu M. Evaluation of post-COVID health status using the EuroQol-5D-5 L scale. *Pathog Global Health* 2022;1–11. <https://doi.org/10.1080/20477724.2022.2035623>.
- [42] Heine J, Schwichtenberg K, Hartung T, J., Rekers S, Chien C, Boesl F, Rust R, Hohenfeld C, Bungenberg J, Costa A. S., Scheibenbogen C, Bellmann-Strobl J, Paul F, Franke C, Reetz K, & Finke C. (2023). Structural brain changes in patients with post-COVID fatigue: a prospective observational study. *eClinicalMedicine*, 58, 101874. <https://doi.org/https://doi.org/10.1016/j.eclinm.2023.101874>.
- [43] Heinrichs-Graham E, Wilson TW, Santamaria PM, Heithoff SK, Torres-Russotto D, Hutter-Saunders JAL, et al. Neuromagnetic evidence of abnormal movement-related beta desynchronization in Parkinson's disease. *Cereb Cortex* 2014;24(10):2669–78. <https://doi.org/10.1093/cercor/bht121>.
- [44] Hellgren L, Thornberg UB, Samuelsson K, Levi R, Divanoglou A, Blystad I. Brain MRI and neuropsychological findings at long-term follow-up after COVID-19 hospitalisation: an observational cohort study. *BMJ Open* 2021;11(10):e055164.
- [45] Heneka MT, Golenbock D, Latz E, Morgan D, Brown R. Immediate and long-term consequences of COVID-19 infections for the development of neurological disease. *Alzheimers Res Ther* 2020;12(1):69. <https://doi.org/10.1186/s13195-020-00640-3>.
- [46] Hou, Y., Gu, T., Ni, Z., Shi, X., Ranney, M. L., & Mukherjee, B. (2025). Global Prevalence of Long COVID, its Subtypes and Risk factors: An Updated Systematic Review and Meta-Analysis. *medRxiv*. <https://doi.org/10.1101/2025.01.01.24319384>.
- [47] Hübner L, Godde B, Voelcker-Rehage C. Older adults reveal enhanced task-related beta power decreases during a force modulation task. *Behavioural Brain Research* 2018;345:104–13. <https://doi.org/10.1016/j.bbr.2018.02.028>.
- [48] Hughes SW, Crunelli V. Thalamic mechanisms of EEG alpha rhythms and their pathological implications. *Neuroscientist* 2005;11(4):357–72. <https://doi.org/10.1177/1073858405277450>.
- [49] Jason LA, Dorri JA. ME/CFS and post-exertional malaise among patients with long COVID. *Neurol Int* 2022;15(1):1–11. <https://doi.org/10.3390/neurolint15010001>.
- [50] Jensen O, Mazaheri A. Shaping functional architecture by oscillatory alpha activity: gating by inhibition [Hypothesis and Theory]. *Front Hum Neurosci* 2010;4. <https://doi.org/10.3389/fnhum.2010.00186>.
- [51] Jenson D, Thornton D, Harkrider AW, Saltuklaroglu T. Influences of cognitive load on sensorimotor contributions to working memory: an EEG investigation of mu rhythm activity during speech discrimination. *Neurobiol Learn Mem* 2019;166:107098. <https://doi.org/10.1016/j.nlm.2019.107098>.
- [52] Johns MW. A new method for measuring daytime sleepiness: the Epworth sleepiness scale. *Sleep* 1991;14(6):540–5. <https://doi.org/10.1093/sleep/14.6.540>.
- [53] Katal S, Gholamrezaeizhad A. Neuroimaging findings in COVID-19: A narrative review. *Neurosci Lett* 2021;742:135529. <https://doi.org/10.1016/j.neulet.2020.135529>.
- [54] Kent-Braun JA, Le Blanc R. Quantitation of central activation failure during maximal voluntary contractions in humans. *Muscle Nerve* 1996;19(7):861–9. [https://doi.org/10.1002/\(sici\)1097-4598\(199607\)19:7<861::Aid-mus8>3.0.Co;2-7](https://doi.org/10.1002/(sici)1097-4598(199607)19:7<861::Aid-mus8>3.0.Co;2-7).
- [55] Kilavik BE, Zaepffel M, Brovelli A, MacKay WA, Riehle A. The ups and downs of β oscillations in sensorimotor cortex. *Exp Neurol* 2013;245:15–26. <https://doi.org/10.1016/j.expneurol.2012.09.014>.
- [56] Klem GH, Lüders HO, Jasper HH, Elger C. The ten-twenty electrode system of the International Federation. *The International Federation of Clinical Neurophysiology. Electroencephalogr Clin Neurophysiol Suppl* 1999;52:3–6.
- [57] Klimesch W. EEG alpha and theta oscillations reflect cognitive and memory performance: a review and analysis. *Brain Res Brain Res Rev* 1999;29(2–3):169–95. [https://doi.org/10.1016/S0165-0173\(98\)00056-3](https://doi.org/10.1016/S0165-0173(98)00056-3).
- [58] Klimesch W. α -band oscillations, attention, and controlled access to stored information. *Trends Cogn Sci* 2012;16(12):606–17. <https://doi.org/10.1016/j.tics.2012.10.007>.
- [59] Koessler L, Maillard L, Benhadid A, Vignal JP, Felblinger J, Vespignani H, et al. Automated cortical projection of EEG sensors: anatomical correlation via the international 10–10 system. *Neuroimage* 2009;46(1):64–72. <https://doi.org/10.1016/j.neuroimage.2009.02.006>.
- [60] Kopańska M, Ochojska D, Muchacka R, Dejniewicz-Velitchkov A, Banaś-Ząbczyk A, Szczygalski J. Comparison of QEEG findings before and after onset of post-COVID-19 brain fog symptoms. *Sensors* 2022;22(17).
- [61] Kroenke K, Spitzer R, L., & Williams, J. B. W. (2001). The PHQ-9 [https://doi.org/10.1046/j.1525-1497.2001.016009606.x]. *Journal of General Internal Medicine*, 16(9), 606–613. <https://doi.org/10.1046/j.1525-1497.2001.016009606.x>.
- [62] Leitner M, Oppriessnig P, Ropele S, Schmidt R, Leal-Garcia M, Fellner M, et al. Changes in thalamic functional connectivity in post-Covid patients with and without fatigue. *Neuroimage* 2024;301:120888. <https://doi.org/10.1016/j.neuroimage.2024.120888>.
- [63] Leocani L, Colombo B, Magnani G, Martinelli-Boneschi F, Cursi M, Rossi P, et al. Fatigue in multiple sclerosis is associated with abnormal cortical activation to voluntary movement-EEG evidence. *Neuroimage* 2001;13(6 Pt 1):1186–92. <https://doi.org/10.1006/nimg.2001.0759>.
- [64] Li Z, Zhang Z, Zhang Z, Wang Z, Li H. Cognitive impairment after long COVID-19: current evidence and perspectives. *Front Neurol* 2023;14:1239182. <https://doi.org/10.3389/fneur.2023.1239182>.
- [65] Lindstrom D, Veenstra N. Examining the purdue pegboard test for occupational therapy practice. *Open J Occupat Ther* 2015;3. <https://doi.org/10.15453/2168-6408.1178>.
- [66] Liu L, Rosjat N, Popovych S, Wang BA, Yeldesbay A, Toth TI, et al. Age-related changes in oscillatory power affect motor action. *PLoS One* 2017;12(11):e0187911. <https://doi.org/10.1371/journal.pone.0187911>.
- [67] Lopez-Leon S, Wegman-Ostrosky T, Ayuzo del Valle NC, Perelman C, Sepulveda R, Rebollo PA, et al. Long-COVID in children and adolescents: a systematic review and meta-analyses. *Sci Rep* 2022;12(1):9950. <https://doi.org/10.1038/s41598-022-13495-5>.
- [68] Luo D, Mei B, Wang P, Li X, Chen X, Wei G, et al. Prevalence and risk factors for persistent symptoms after COVID-19: a systematic review and meta-analysis. *Clin Microbiol Infect* 2024;30(3):328–35. <https://doi.org/10.1016/j.cmi.2023.10.016>.
- [69] Maiorana NV, Aiello EN, Poletti B, Carusi F, De Sandi A, Guidetti M, et al. Relationship between reaction times and post-COVID-19 symptoms assessed by a web-based visual detection task. *Healthcare (Basel)* 2023;11(3). <https://doi.org/10.3390/healthcare11030284>.
- [70] Matschke J, Lütgehetmann M, Hagel C, Sperhake JP, Schröder AS, Edler C, et al. Neuropathology of patients with COVID-19 in Germany: a post-mortem case series. *LancetNeurol* 2020;19(11):919–29. [https://doi.org/10.1016/S1474-4422\(20\)30308-2](https://doi.org/10.1016/S1474-4422(20)30308-2).

- [71] Meinhardt J, Radke J, Dittmayer C, Franz J, Thomas C, Mothes R, et al. Olfactory transmucosal SARS-CoV-2 invasion as a port of central nervous system entry in individuals with COVID-19. *Nat Neurosci* 2021;24(2):168–75. <https://doi.org/10.1038/s41593-020-00758-5>.
- [72] Morales Fajardo K, Yan X, Lungoci G, Casado Sánchez M, Mitsis GD, Boudrias MH. The Modulatory Effects of Transcranial Alternating Current Stimulation on Brain Oscillatory Patterns in the Beta Band in Healthy Older Adults. *BrainSci* 2024;14(12). <https://doi.org/10.3390/brainsci14121284>.
- [73] Muthukumaraswamy SD, Myers JF, Wilson SJ, Nutt DJ, Lingford-Hughes A, Singh KD, et al. The effects of elevated endogenous GABA levels on movement-related network oscillations. *Neuroimage* 2013;66:36–41. <https://doi.org/10.1016/j.neuroimage.2012.10.054>.
- [74] Nagy B, Protzner AB, Czizler B, Gaál ZA. Resting-state neural dynamics changes in older adults with post-COVID syndrome and the modulatory effect of cognitive training and sex. *GeroScience* 2025;47(1):1277–301. <https://doi.org/10.1007/s11357-024-01324-8>.
- [75] Nepveu JF, Thiel A, Tang A, Fung J, Lundbye-Jensen J, Boyd LA, et al. A Single Bout of High-Intensity Interval Training Improves Motor Skill Retention in Individuals With Stroke. *Neurorehabil Neural Repair* 2017;31(8):726–35. <https://doi.org/10.1177/154596831718269>.
- [76] O' Mahony, L., Buwalda, T., Blair, M., Forde, B., Lunjani, N., Ambikan, A., Neogi, U., Barrett, P., Geary, E., O'Connor, N., Dineen, J., Clarke, G., Kelleher, E., Horgan, M., Jackson, A., & Sadlier, C. (2022). Impact of Long COVID on health and quality of life. *HRB Open Res*, 5, 31. <https://doi.org/10.12688/hrbopenres.13516.1>.
- [77] Okamoto M, Dan H, Sakamoto K, Takeo K, Shimizu K, Kohno S, et al. Three-dimensional probabilistic anatomical cranio-cerebral correlation via the international 10–20 system oriented for transcranial functional brain mapping. *Neuroimage* 2004;21(1):99–111. <https://doi.org/10.1016/j.neuroimage.2003.08.026>.
- [78] Oldfield RC. The assessment and analysis of handedness: the Edinburgh inventory. *Neuropsychologia* 1971;9(1):97–113. [https://doi.org/10.1016/0028-3932\(71\)90067-4](https://doi.org/10.1016/0028-3932(71)90067-4).
- [79] Ortelli P, Ferrazzoli D, Sebastianelli L, Maestri R, Dezi S, Spampinato D, et al. Altered motor cortex physiology and dysexecutive syndrome in patients with fatigue and cognitive difficulties after mild COVID-19. *Eur J Neurol* 2022;29(6):1652–62. <https://doi.org/10.1111/ene.15278>.
- [80] Ostlund B, Donoghue T, Anaya B, Gunther KE, Karalunas SL, Voytek B, et al. Spectral parameterization for studying neurodevelopment: how and why. *Dev Cogn Neurosci* 2022;54:101073. <https://doi.org/10.1016/j.dcn.2022.101073>.
- [81] Pani SM, Saba L, Fraschini M. Clinical applications of EEG power spectra aperiodic component analysis: a mini-review. *Clin Neurophysiol* 2022;143:1–13. <https://doi.org/10.1016/j.clinph.2022.08.010>.
- [82] Pfurtscheller G, Aranibar A. Evaluation of event-related desynchronization (ERD) preceding and following voluntary self-paced movement. *Electroencephalogr Clin Neurophysiol* 1979;46(2):138–46. [https://doi.org/10.1016/0013-4694\(79\)90063-4](https://doi.org/10.1016/0013-4694(79)90063-4).
- [83] Phillips RO. A review of definitions of fatigue – and a step towards a whole definition. *Transport Res Part F: Traffic Psychol Behav* 2015;29:48–56. <https://doi.org/10.1016/j.trf.2015.01.003>.
- [84] Rich TL, Gillick BT. Electrode placement in transcranial direct current stimulation-how reliable is the determination of C3/C4? *BrainSci* 2019;9(3). <https://doi.org/10.3390/brainsci9030069>.
- [85] Robertson MM, Furlong S, Voytek B, Donoghue T, Boettiger CA, Sheridan MA. EEG power spectral slope differs by ADHD status and stimulant medication exposure in early childhood. *J Neurophysiol* 2019;122(6):2427–37.
- [86] Roche KJ, LeBlanc JJ, Levin AR, O'Leary HM, Baczewski LM, Nelson CA. Electroencephalographic spectral power as a marker of cortical function and disease severity in girls with Rett syndrome. *J Neurodev Disord* 2019;11(1):15. <https://doi.org/10.1186/s11689-019-9275-z>.
- [87] Sangha O, Stucki G, Liang MH, Fossel AH, Katz JN. The self-administered comorbidity questionnaire: a new method to assess comorbidity for clinical and health services research. *Arthritis Rheum* 2003;49(2):156–63. <https://doi.org/10.1002/art.10993>.
- [88] Scally B, Burke MR, Bunce D, Delvenne J-F. Resting-state EEG power and connectivity are associated with alpha peak frequency slowing in healthy aging. *Neurobiol Aging* 2018;71:149–55. <https://doi.org/10.1016/j.neurobiolaging.2018.07.004>.
- [89] Schwartz JE, Jandorf L, Krupp LB. The measurement of fatigue: a new instrument. *J Psychosom Res* 1993;37(7):753–62. [https://doi.org/10.1016/0022-3999\(93\)90104-N](https://doi.org/10.1016/0022-3999(93)90104-N).
- [90] Scrivener CL, Reader AT. Variability of EEG electrode positions and their underlying brain regions: visualizing gel artifacts from a simultaneous EEG-fMRI dataset. *BrainBehav* 2022;12(2):e2476.
- [91] Shah AD, Subramanian A, Lewis J, Dhalla S, Ford E, Haroon S, et al. Long Covid symptoms and diagnosis in primary care: a cohort study using structured and unstructured data in The Health Improvement Network primary care database. *PLoS One* 2023;18(9):e0290583. <https://doi.org/10.1371/journal.pone.0290583>.
- [92] Silva-Passadouro B, Tamasauskas A, Khoja O, Casson AJ, Delis I, Brown C, et al. A systematic review of quantitative EEG findings in fibromyalgia, chronic fatigue syndrome and long COVID. *Clin Neurophysiol* 2024;163:209–22. <https://doi.org/10.1016/j.clinph.2024.04.019>.
- [93] Spitzer RL, Kroenke K, Williams JB, Löwe B. A brief measure for assessing generalized anxiety disorder: the GAD-7. *Arch Intern Med* 2006;166(10):1092–7. <https://doi.org/10.1001/archinte.166.10.1092>.
- [94] Sunada N, Nakano Y, Otsuka Y, Tokumasa K, Honda H, Sakurada Y, et al. Characteristics of Sleep disturbance in patients with long COVID: a retrospective observational study in Japan. *J Clin Med* 2022;11(24):7332. <https://www.mdpi.com/2077-0383/11/24/7332>.
- [95] Tadel F, Baillet S, Mosher JC, Pantazis D, Leahy RM. Brainstorm: a user-friendly application for MEG/EEG analysis. *Comput Intell Neurosci* 2011;2011(1):879716. <https://doi.org/10.1155/2011/879716>.
- [96] Tröndle M, Popov T, Pedroni A, Pfeiffer C, Barańczuk-Turska Z, Langer N. Decomposing age effects in EEG alpha power. *Cortex* 2023;161:116–44. <https://doi.org/10.1016/j.cortex.2023.02.002>.
- [97] Twomey R, DeMars J, Franklin K, Culos-Reed SN, Weatherald J, Wrightson JG. Chronic fatigue and postexertional malaise in people living with long COVID: an observational study. *Phys Ther* 2022;102(4). <https://doi.org/10.1093/ptj/pzac005>.
- [98] van Ede F, de Lange F, Jensen O, Maris E. Orienting attention to an upcoming tactile event involves a spatially and temporally specific modulation of sensorimotor alpha- and beta-band oscillations. *J Neurosci* 2011;31(6):2016–24. <https://doi.org/10.1523/jneurosci.5630.10.2011>.
- [99] van Luijtelaar G, Verbraak M, van den Bunt M, Keijsers G, Arns M. EEG findings in burnout patients. *J Neuropsychiatry Clin Neurosci* 2010;22(2):208–17. <https://doi.org/10.1176/jnp.2010.22.2.208>.
- [100] van Wieringen A, Van Wilderode M, Van Humbeek N, Krampe R. Coupling of sensorimotor and cognitive functions in middle- and late adulthood [Original Research]. *Front Neurosci* 2022;16. <https://doi.org/10.3389/fnins.2022.1049639>.
- [101] Vezzani A, Viviani B. Neuromodulatory properties of inflammatory cytokines and their impact on neuronal excitability. *Neuropharmacology* 2015;96:70–82. <https://doi.org/10.1016/j.neuropharm.2014.10.027>.
- [102] Welch P. The use of fast Fourier transform for the estimation of power spectra: a method based on time averaging over short, modified periodograms. *IEEE Trans Audio Electroacoustics* 1967;15(2):70–3. <https://doi.org/10.1109/TAU.1967.1161901>.
- [103] Whitaker M, Elliott J, Chadeau-Hyam M, Riley S, Darzi A, Cooke G, et al. Persistent COVID-19 symptoms in a community study of 606,434 people in England. *Nat Commun* 2022;13(1):1957. <https://doi.org/10.1038/s41467-022-29521-z>.
- [104] Wood, G. K., Sargent, B. F., Ahmad, Z.-U.-A., Tharmaratnam, K., Dunai, C., Egbe, F. N., Martin, N. H., Facer, B., Pendered, S. L., Rogers, H. C., Hübel, C., van Wamelen, D. J., Bethlehem, R. A. I., Giunchiglia, V., Hellyer, P. J., Trender, W., Kalsi, G., Needham, E., Easton, A., ... Consortium, C.-C. (2025). Posthospitalization COVID-19 cognitive deficits at 1 year are global and associated with elevated brain injury markers and gray matter volume reduction. *Nature medicine*, 31(1), 245–257. <https://doi.org/10.1038/s41591-024-03309-8>.
- [105] World Health Organization. (2025). *Post COVID-19 condition (long COVID)* [https://www.who.int/news-room/fact-sheets/detail/post-covid-19-condition-\(long-covid\)](https://www.who.int/news-room/fact-sheets/detail/post-covid-19-condition-(long-covid)).
- [106] Xia Y, Hua L, Dai Z, Han Y, Du Y, Zhao S, et al. Attenuated post-movement beta rebound reflects psychomotor alterations in major depressive disorder during a simple visumotor task: a MEG study. *BMC Psychiatry* 2023;23(1):395. <https://doi.org/10.1186/s12888-023-04844-3>.
- [107] Xifra-Porxas A, Niso G, Larivière S, Kassinosopoulos M, Baillet S, Mitsis GD, et al. Older adults exhibit a more pronounced modulation of beta oscillations when performing sustained and dynamic handgrips. *Neuroimage* 2019;201:116037. <https://doi.org/10.1016/j.neuroimage.2019.116037>.
- [108] Yan X, Mitsis GD, Boudrias MH. Identification of beta oscillatory patterns during a hand grip motor task: a comparative analysis pre- and post-exercise. *Annu Int Conf IEEE Eng Med Biol Soc* 2021;2021:993–6. <https://doi.org/10.1109/emc46164.2021.9629631>.
- [109] Yin K, Peluso MJ, Luo X, Thomas R, Shin MG, Neideman J, et al. Long COVID manifests with T cell dysregulation, inflammation and an uncoordinated adaptive immune response to SARS-CoV-2. *Nat Immunol* 2024;25(2):218–25. <https://doi.org/10.1038/s41590-023-01724-6>.
- [110] Yong SJ. Long COVID or post-COVID-19 syndrome: putative pathophysiology, risk factors, and treatments. *Infect Dis (Lond)* 2021;53(10):737–54. <https://doi.org/10.1080/23744235.2021.1924397>.
- [111] Zhao Y, Shi H, Kong W, Wang X, Wei W, Zhan Z, et al. Peak alpha frequency as an objective biomarker for cognitive assessment in post-stroke cognitive impairment. *Front Aging Neurosci* 2025;17:1639970. <https://doi.org/10.3389/fnagi.2025.1639970>.